A verified empirical separator for the skipped curves

29 October 2026. A concrete criterion now matches all 8 published and 11 omitted curves in the four fields with both labels: **published curves have fewer strictly split small primes up to 691 than every omitted curve in the same field**. The two additional published P-384/P-521 controls are retained, with no same-field omissions available to test discrimination. This independently reproduces an externally reported pattern. Historical use and either vulnerability hypothesis remain unproved.

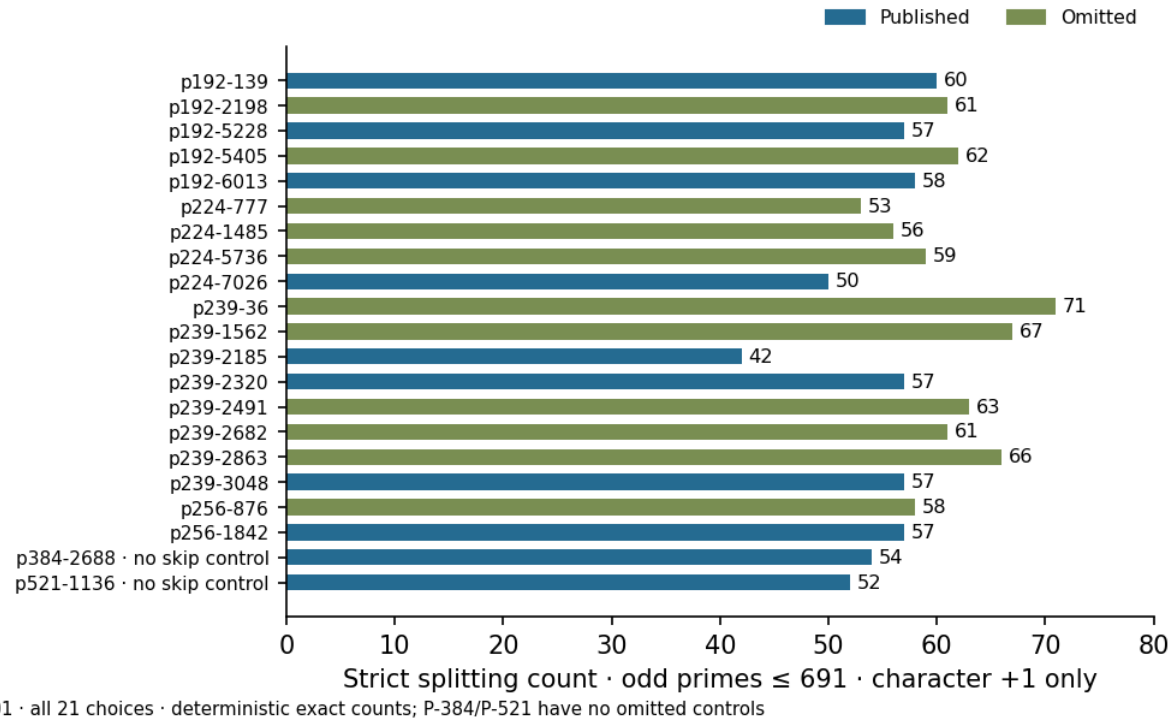
The criterion and measured values

Counts, degree cutoffs and rule thresholds below are decimal.

Define $t=p+1-n$, $\Delta=t^2-4p$, and $E691 = \#\{\text{odd primes } \ell \le 691 : (\Delta/\ell)=+1\}$. Accept a candidate in the compared fields when $E691 \le 60$ for P-192, ≤ 50 for P-224, ≤ 57 for P-239 or ≤ 57 for P-256. These per-field bounds describe the observed published envelopes; they are not authenticated original acceptance constants.

Field	Verified published / omitted counts
P-192	Published 60,57,58; omitted 61,62
P-224	Published 50; omitted 53,56,59
P-239	Published 42,57,57; omitted 71,67,63,61,66
P-256	Published 57; omitted 58
P-384	Published 54; no omitted same-field control
P-521	Published 52; no omitted same-field control

Every count is independently checked in the 21-row table. The values are exact on this census, with no sampling interval. The criterion uses separate field envelopes. A global threshold cannot work: P-192's published count 60 exceeds P-224's omitted count 53.



Attempt 001 · all 21 choices · deterministic exact counts; P-384/P-521 have no omitted controls

What was searched and what the calibration means

The asanso post of 17 October reported this pattern before this study. We read the authored text through that mirror; the direct X endpoint returned 403. The reported outcome and all labels were visible before freeze, so this is neither a new discovery nor an independent historical holdout.

The native programs retain every one of 303 prime cutoff positions through 2,000. The boundary at 2 contributes zero; its character is explicitly NA_ell2. Only the 302 odd primes contribute characters. All 6,342 odd-prime character cells for the 21 curves are recomputed. For each cutoff, the fixed lower-count direction must separate every same-field published/omitted pair. The four successful cutoffs are **691, 701, 709 and 769**. No different direction or later chosen feature is silently included.

All 4,480 label assignments preserving published counts in each original stream are enumerated. Exactly **23/4,480** admit some separating cutoff in the stated strict-count family, matching the externally reported count. Exactly one labeling separates at the fixed value 691, but that cutoff was selected retrospectively in the earlier work. The smaller fixed-cutoff fraction must not be presented as a new confirmatory result. The 303-position enumeration accounts for that cutoff search only. It does not account for every other trait family, direction or hypothesis previously investigated, and does not establish exchangeability of historical selection decisions.

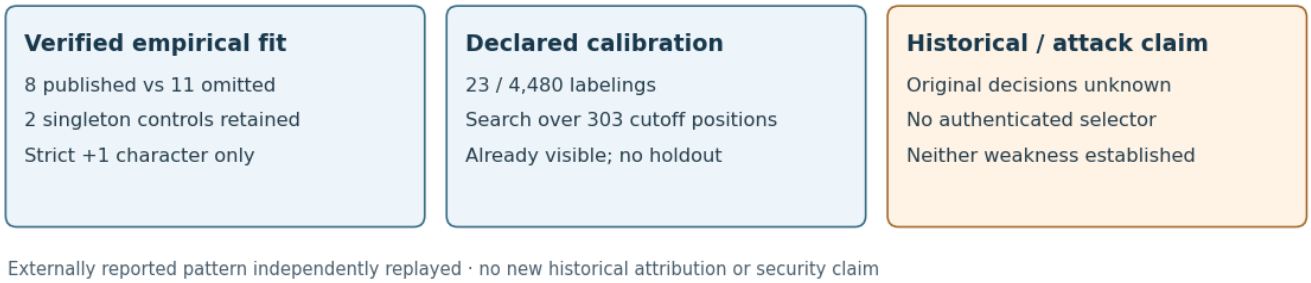
The earlier complete-factor study already has empirical two-test and three-test fits. This structured count criterion is another empirical fit, with the stated narrower search calibration, not proof of original use or of deliberate selection. No independent acceptance/visit trace or original generation program has been found or supplied; the user has none.

Ramification changes the answer

The statistic deliberately counts character +1, excluding zero. Its name here is **strict splitting count**. Some uses of “Elkies prime” also include ramified primes; these conventions must remain separate. A positive discriminant character gives distinct Frobenius eigenvalues modulo an odd prime and stable cyclic subgroups; ramified cases are not a lack of rational isogenies and are not simply discarded as mathematically invalid.

At 691, P-192 index 139 has strict count 60 and three ramified primes, making its inclusive count 63. Omitted index 2198 has inclusive count 61 and index 5405 has 62. Thus an inclusive upper envelope accepting every publication also keeps these omissions. **The cutoff-691 separation fails under that broader definition.** The separate inclusive cutoff-search family has 20 successful labelings; it is a sensitivity result, not a replacement chosen to strengthen the strict-family claim.

The old native snapshots included ramification and stopped their fixed count snapshots at 499. The older product-coverage proxy, which multiplied eligible primes until a Hasse threshold was exceeded, is another quantity. Its earlier counterexamples remain correct; this result does not overwrite those records or change their definition after observing an outcome.



Independent verification and actual execution

The GMP producer enumerates quadratic residues for trial-divided small primes, computes every count and scans the full stream-matched label space. The separate PARI verifier uses its prime routine and Kronecker symbols, direct same-field pair comparisons and recursive label combinations. It reconstructs and compares all six output files, including the complete headline summary. Underlying GMP integer primitives may be shared; this is not a claim of fully independent underlying big-integer libraries. Input orders are reused from the earlier certified cohort; there is no new point count or input-primality proof in this replay.

Actual source at launch was 077c41e7d414cf72faf20bf9b474b7c35dae7bd8. The canonical protocol was frozen and committed before real computation, with digest eeb7482bcda69477a360a7c28749b7f737d377dbe3298bec94346e4efbcdbbb7. The exact source/binaries come from preparation commit 043a568661ebf1c40a4ead333889d814c32d045c. All 21 inputs, six streams, both singleton published controls and the exceptional character at 2 remain.

Execution flow	Observed outcome
Six phases planned / attempted / completed	6 / 6 / 6; no resume, interruption or exclusion
Positive control	Native exit 0; residue catalogs, planted separator, nonseparator and toy label count pass
Production and independent verification	Native exits 0 and 0; all 21 rows, 6,342 characters, 303 cutoff positions and 4,480 assignments checked
Corrupted character, calibration and summary	Native exits 1,1,1 as required; each reports the intended mismatching file
Canonical export admission	File verification passes with zero findings

The three negative exits are correct refusals, not successful verification of corrupted data. Their original outputs and actual exit 1 files are retained. The positive limit was 30 seconds; production/verifier 120 seconds each; each refusal 30 seconds; all native phases used 1536 MiB virtual address space. The uploaded checkpoint gate passed before the run. No automatic retry occurred.

The run uses the declared local exact-analysis fallback: no EKS MCP is installed; the unchanged portable request validator only accepts ecbench, and existing CodeBuild project inspection was denied. No paid job or policy change was substituted. Actual computation admits no performance clock, isolation tier or speedup. The source-controlled planning self-test observations are separate. Their first attempt failed before a control ran because /usr/bin/time was unavailable; the second used Bash's time keyword and passed. Both attempts remain preserved.

See canonical export, verification, run report and planning report. Raw custody and hydration are documented in REPRODUCE.md.

Connection to the base point and b

This statistic depends on p and the certified order n, through the trace. It does not depend on choosing a different generator on the same curve. The two P-224 coefficient representatives used earlier preserve that order, so this profile also avoids the coordinate/root-convention dependence of some earlier fitted rules. This is a structural difference in the measured curves, not a recovered relation between unknown historical skipped base points and b. All eleven such historical base points remain unavailable.

The earlier G–b census retains 100,548 verified point-trait cells and 147,875 same-field difference cells. Its 19,152 single-envelope tests and 880 scalar tests have zero complete separators or scalar matches. Those outcomes remain unchanged. The assigned $x=0/x=1$ point relations $y^2=b/y^2=b-2$ are forced by the equation. The earlier FreeLIP RNG relation to six published base points is separate prior work; this new profile neither reconstructs nor refutes it.

Status against the full request

Requirement	Evidence and remaining gap
Complete four unresolved factorizations	Previously proved complete in the 84-record full-21 follow-up
Find a criterion matching the skips	Verified empirical strict-count envelope matches all 11 omissions against all 8 paired publications; two singleton publications retained
Establish the historical explanation	Unresolved: no authenticated caller/decision trace or independent historical predictions
Measure curve differences	Complete for the stated inherited, point-trait and strict-character families; not an enumeration of all possible mathematical functions
Explain G versus b	Defining-equation relations and fixed scalar recipes checked; historical omitted generators remain unknown
Assess weak selected / weak skipped	Neither vulnerability is established; no DLP or applicable attack construction
PDF and PR	This current PDF, supporting reports, and existing PR 2666

There are unboundedly many functions of curve parameters; “all possible differences” is not a finite measurement specification. This report retains the wider request as open beyond the declared families. The new observed separator is concrete and reproducible. Establishing why the original program made those decisions still requires an authenticated mechanism or independent historical predictions. More strict splitting primes alone do not establish weaker ECDLP security or a useful attack route.

Methods and limitations were reviewed against Shparlinski–Sutherland, arXiv1112.3390v2, especially the odd-prime definition and averaged distribution scope, the externally reported profile, the older SEA replay, the complete-factor study, and the historical-source audit. No explicit isogeny map or new algebraic identity is claimed; the result is the independently checked finite integer census. Existing factor, embedding, G–b and IC/performance graphs retain their earlier values. The new count graph and diagram show the added evidence without promoting an attack, security or historical-use claim.